# Principles of Anatomy and Physiology

**Gerard Tortora and Bryan Derrickson**

Sixteenth Edition

## Chapter 1

An Introduction to the Human Body

# Introduction

The purpose of the chapter is to:

- Introduce the disciplines of anatomy and physiology
- Discuss the organization of the human body
- Reveal shared properties of all living things
- Discuss the concept of homeostasis

# Anatomy & Physiology Defined

# Anatomy vs. Physiology

- Anatomy is the study of structure
- Physiology is the study of function

# Anatomy vs. Physiology

## **Table 1.1** Selected Branches of Anatomy and Physiology

| Branch of Anatomy | Study of |
|---|---|
| **Developmental biology** | The growth and development of an individual from fertilization to death. |
| **Embryology** (em′-brē-OL-ō-jē; *embry-* = embryo; *-logy* = study of) | The first eight weeks of growth and development after fertilization of a human egg; the earliest stage of developmental biology. |
| **Cell biology** | Cellular structure and functions. |
| **Histology** (his-TOL-ō -jē; *hist-* = tissue) | Microscopic structure of tissues. |
| **Gross anatomy** | Structures that can be examined without a microscope. |

# Anatomy vs. Physiology

| Branch of Anatomy | Study of |
|---|---|
| **Systemic anatomy** | Structure of specific systems of the body such as the nervous or respiratory systems. |
| **Regional anatomy** | Specific regions of the body such as the head or chest. |
| **Surface (topographical) anatomy** | Surface markings of the body to understand internal anatomy through visualization and palpation (gentle touch). |
| **Imaging anatomy** | Internal body structures that can be visualized with techniques such as x-rays, MRI, CT scans, and other technologies for clinical analysis and medical intervention. |
| **Clinical anatomy** | The application of anatomy to the practice of medicine, dentistry, and other health-related sciences, for example, to aid in the diagnosis and treatment of disease. |
| **Pathological anatomy** (path'-ō-LOJ-i-kal; *path-* = disease) | Structural changes (gross to microscopic) associated with disease. |

# Anatomy vs. Physiology

| Branch of Physiology | Study of |
|---|---|
| **Molecular physiology** | Functions of individual molecules such as proteins and DNA. |
| **Neurophysiology** (NOOR-ō-fiz-ē-ol'-ō-jē; *neuro-* = nerve) | Functional properties of nerve cells. |
| **Endocrinology** (en'-dō-kri-NOL-ō-jē; *endo-* = within; *-crin* = secretion) | Hormones (chemical regulators in the blood) and how they control body functions. |
| **Cardiovascular physiology** (kar-dē-ō-VAS-kū-lar; *cardi-* = heart; *vascular* = blood vessels) | Functions of the heart and blood vessels. |

# Anatomy vs. Physiology

| Branch of Physiology | Study of |
|---|---|
| **Immunology** (im'-ū-NOL-ō-jē; *immun-* = not susceptible) | The body's defenses against disease-causing agents. |
| **Respiratory physiology** (RES-pi-ra-tōr-ē; *respira-* = to breathe) | Functions of the air passageways and lungs. |
| **Renal physiology** (RĒ-nal; *ren-* = kidney) | Functions of the kidneys. |
| **Exercise physiology** | Changes in cell and organ functions due to muscular activity. |
| **Pathophysiology** (Path-ō-fiz-ē-ol'-ō-jē) | Functional changes associated with disease and aging. |

# Levels of Structural Organization and Body Systems

# Levels of Structural Organization

Chemical: atoms and molecules

Cellular: cells and their organelles

Tissue: group of similar cells

Organ: contains two or more types of tissues

Organ system: organs that work closely together

Organismal: all organ systems of a complete individual




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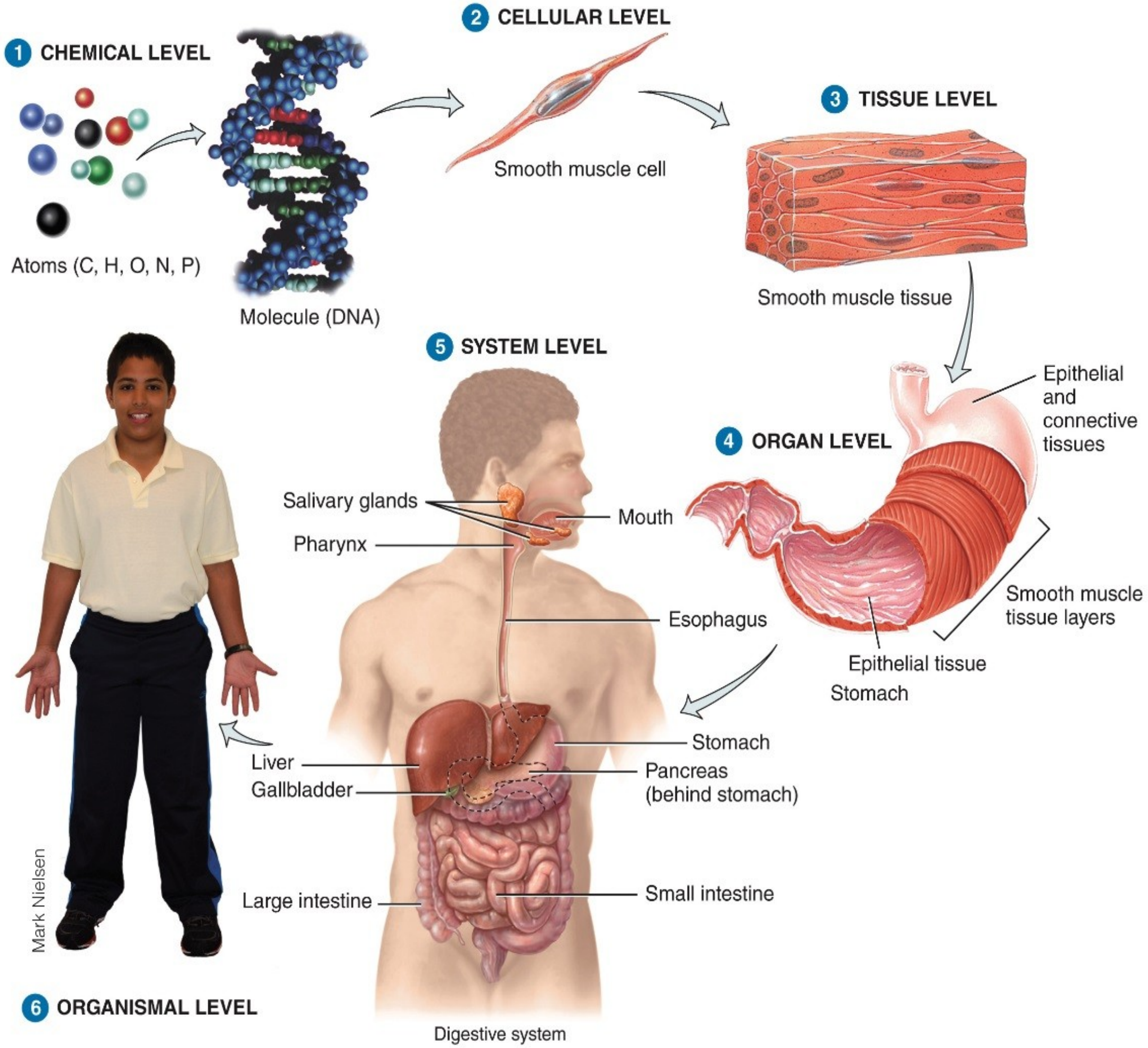
# Levels of Structural Organization



# Systems of the Human Body

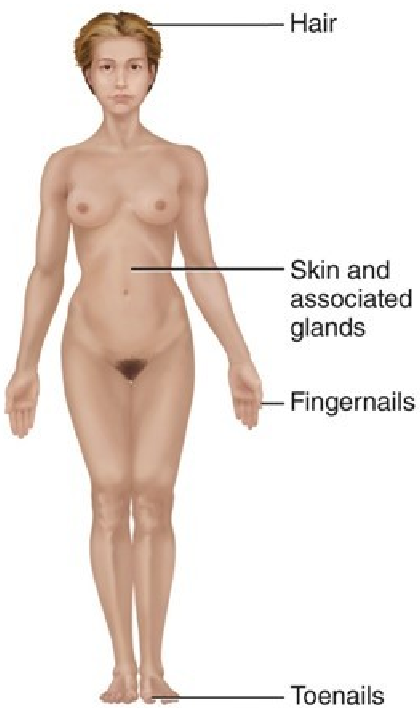
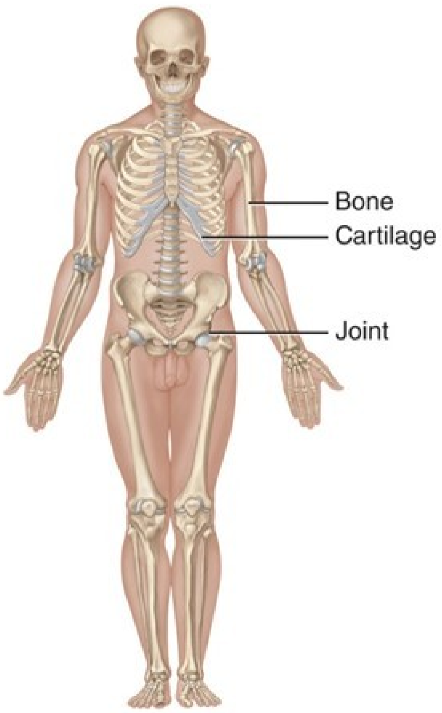
The eleven systems of the human body are the integumentary, skeletal, muscular, nervous, endocrine, cardiovascular, lymphatic/immune, respiratory, digestive, urinary, and reproductive

# Systems of the Human Body

## **Table 1.2** The Eleven Systems of the Human Body

| Integumentary System (chapter 5) | Skeletal System (chapters 6-9) |
|---|---|
| ***Components***: **Skin** and associated structures, such as **hair, fingernails** and **toenails, sweat glands,** and **oil glands.** | ***Components***: **Bones** and **joints** of the body and their associated **cartilages**. |
| ***Functions***: Protects body; helps regulate body temperature; eliminates some wastes; helps make vitamin D; detects sensations such as touch, pain, warmth, and cold; stores fat and provides insulation. | ***Functions***: Supports and protects body; provides surface area for muscle attachments; aids body movements; houses cells that produce blood cells; stores minerals and lipids (fats). |

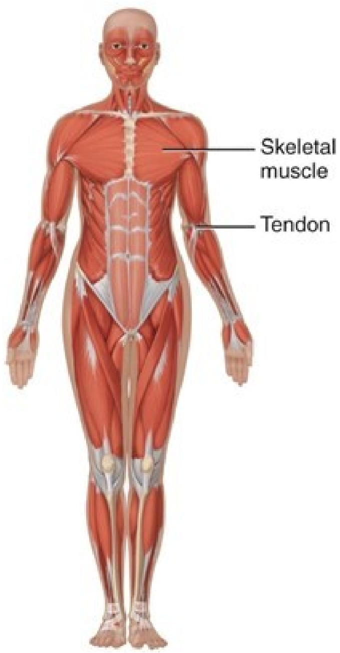
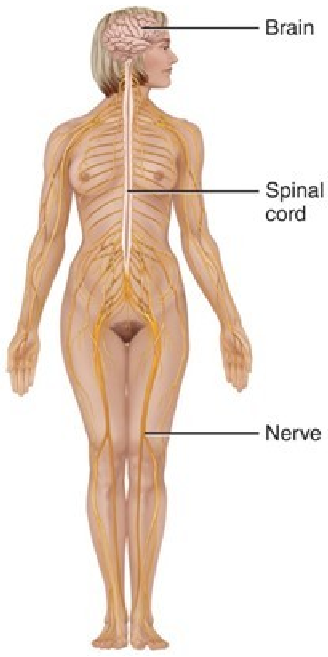
# Systems of the Human Body

| Integumentary System (chapter 5) | Skeletal System (chapters 6-9) |
| --- | --- |
|  |  |

# Systems of the Human Body

| Muscular System (Chapters 10, 11) | Nervous System (Chapters 12-17) |
|---|---|
| ***Components***: Specifically, **skeletal muscle tissue**—muscle usually attached to bones (other muscle tissues include smooth and cardiac). | ***Components***: **Brain, spinal cord, nerves,** and special sense organs, such as **eyes** and **ears.** |
| ***Functions***: Participates in body movements, such as walking; maintains posture; produces heat. | ***Functions***: Generates action potentials (nerve impulses) to regulate body activities; detects changes in body's internal and external environments, interprets changes, and responds by causing muscular contractions or glandular secretions. |

# Systems of the Human Body

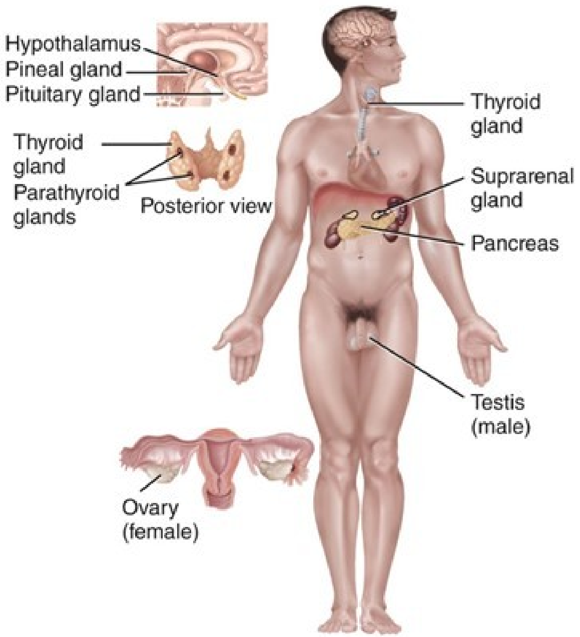
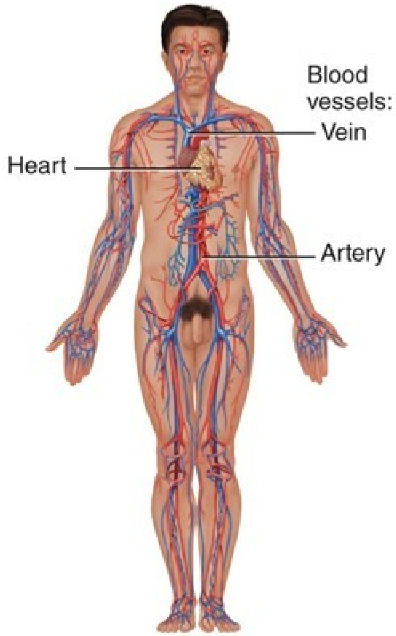
| Muscular System (Chapters 10, 11) | Nervous System (Chapters 12-17) |
| --- | --- |
|  |  |

# Systems of the Human Body

## **Table 1.2** The Eleven Systems of the Human Body

| Endocrine System (chapter 18) | Cardiovascular System (chapters 19-21) |
|---|---|
| ***Components***: Hormone-producing glands **(pineal gland, hypothalamus, pituitary gland, thymus, thyroid gland, parathyroid glands, adrenal glands, pancreas, ovaries,** and **testes**) and hormone-producing cells in several other organs. | ***Components*: Blood, heart,** and **blood vessels.** |
| ***Functions*** : Regulates body activities by releasing hormones (chemical messengers transported in blood from endocrine gland or tissue to target organ). | ***Functions***: Heart pumps blood through blood vessels; blood carries oxygen and nutrients to cells and carbon dioxide and wastes away from cells and helps regulate acid–base balance, temperature, and water content of body fluids; blood components help defend against disease and repair damaged blood vessels. |

# Systems of the Human Body

| Endocrine System (chapter 18) | Cardiovascular System (chapters 19-21) |
|---|---|
|  |  |

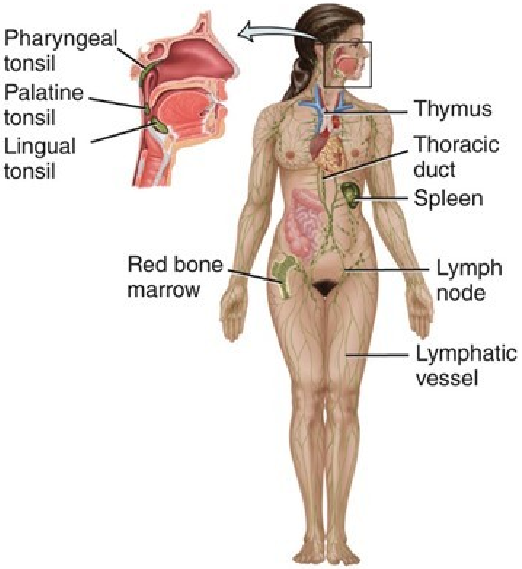
# Systems of the Human Body

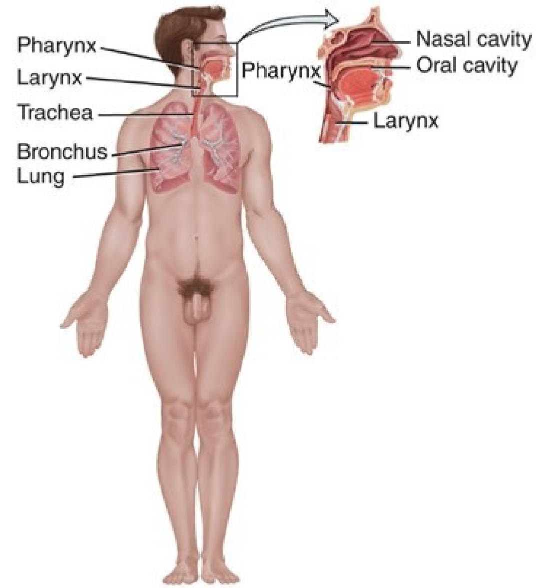
## **Table 1.2** The Eleven Systems of the Human Body

| Lymphoid (Lymphatic) System and Immunity (Chapter 22) | Respiratory System (Chapter 23) |
|---|---|
| ***Components***: **Lymphatic fluid (lymph plasma) and lymphatic vessels; spleen, thymus, lymph nodes, and tonsils;** cells that carry out immune responses (**B cells**, **T cells**, and others). | ***Components***: **Lungs** and air passageways such as the **pharynx** (throat), **larynx** (voice box), **trachea** (windpipe), and **bronchial tubes** leading into and out of lungs. |
| ***Functions***: Returns proteins and fluid to blood; carries lipids from gastrointestinal tract to blood; contains sites of maturation and proliferation of B cells and T cells that protect against disease-causing microbes. | ***Functions***: Transfers oxygen from inhaled air to blood and carbon dioxide from blood to exhaled air; helps regulate acid– base balance of body fluids; air flowing out of lungs through vocal cords produces sounds. |

# Systems of the Human Body

| Lymphatic System and Immunity (chapter 22) | Respiratory System (chapter 23) |
|---|---|

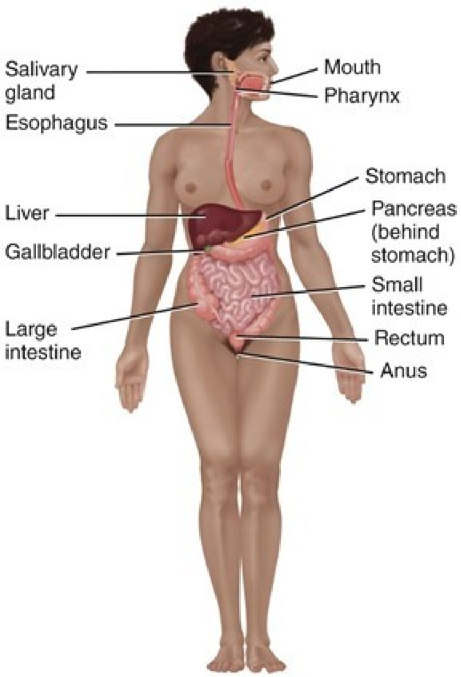
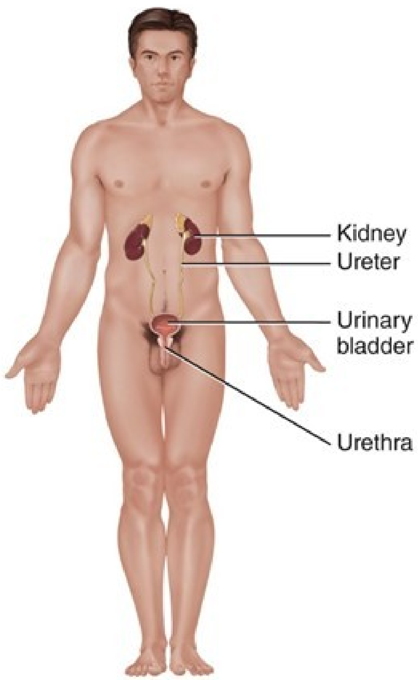


# Systems of the Human Body

| Digestive System (chapter 24) | Urinary System (chapter 26) |
|---|---|
| ***Components***: Organs of gastrointestinal tract, a long tube that includes the **mouth, pharynx** (throat), **esophagus** (food tube), **stomach**, **small** and **large intestines**, and **anus**; also includes accessory organs that assist in digestive processes, such as **salivary glands, liver, gallbladder,** and **pancreas**. | ***Components*****: Kidneys, ureters, urinary bladder**, and **urethra**. |
| ***Functions***: Achieves physical and chemical breakdown of food; absorbs nutrients; eliminates solid wastes. | ***Functions***: Produces, stores, and eliminates urine; eliminates wastes and regulates volume and chemical composition of blood; helps maintain the acid–base balance of body fluids; maintains body's mineral balance; helps regulate production of red blood cells. |

# Systems of the Human Body

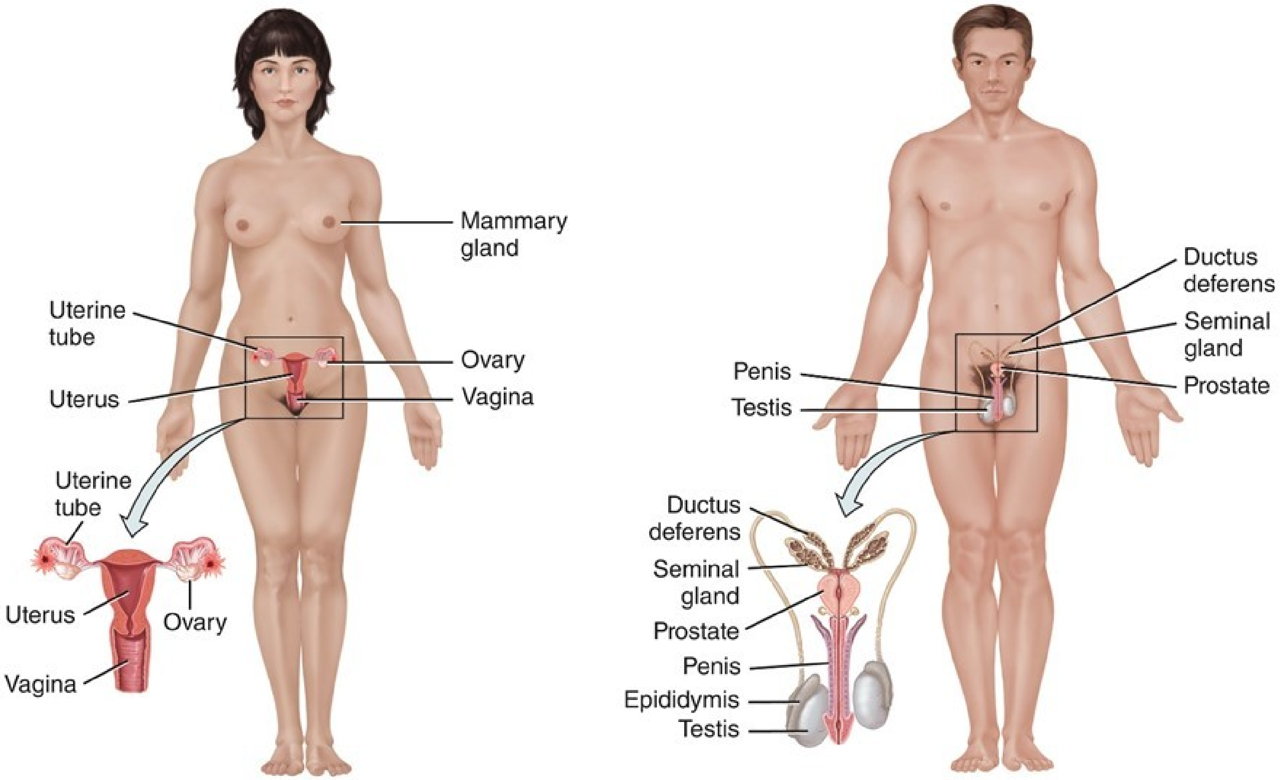
| Digestive System (chapter 24) | Urinary System (chapter 26) |
|---|---|
|  |  |

# Systems of the Human Body

| Genital (Reproductive) Systems (Chapter 28) |
|---|
| ***Components***: **Gonads** (**testes** in males and **ovaries** in females) and associated organs (**uterine tubes**, **uterus**, **vagina**, **clitoris**, and **mammary glands** in females and **epididymis**, **ductus deferens**, **seminal glands**, **prostate**, and **penis** in males). |
| ***Functions***: Gonads produce gametes (sperm or oocytes) that unite to form a new organism; gonads also release hormones that regulate reproduction and other body processes; associated organs transport and store gametes; mammary glands produce milk |

# Systems of the Human Body

## Reproductive Systems (chapter 28)



# Characteristics of the Living Human Organism

# Basic Life Processes

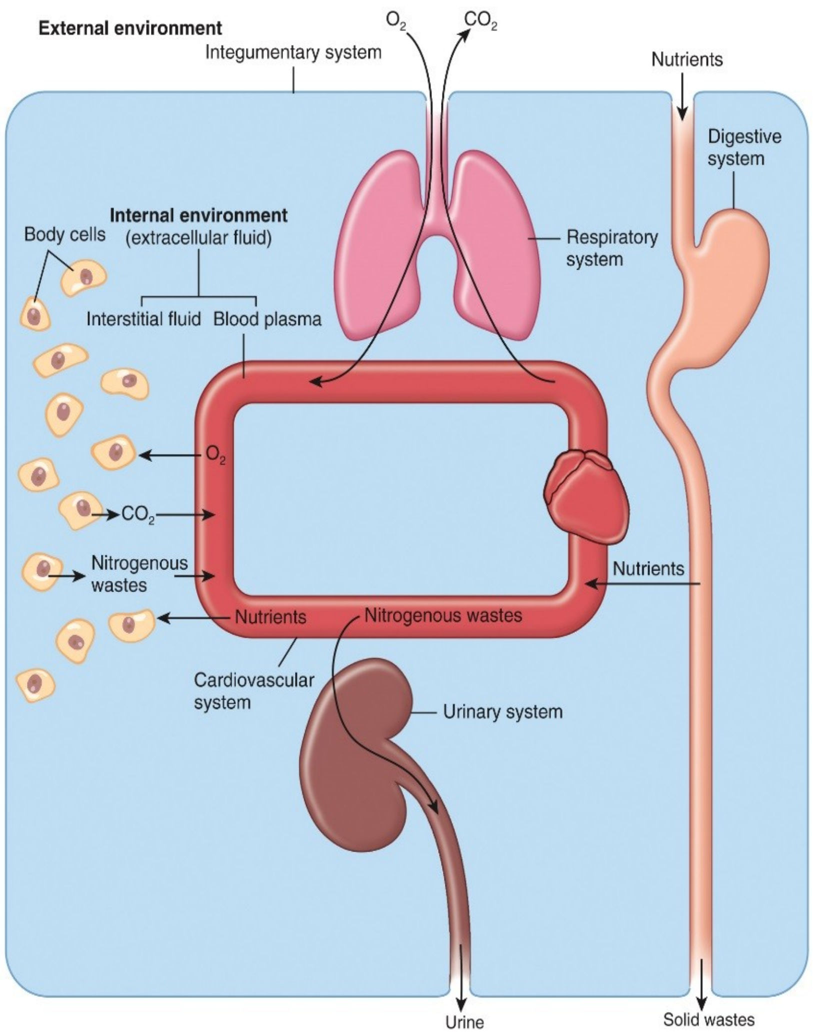
- All living things have certain characteristics that distinguish them from nonliving things
- Life processes in humans include metabolism, responsiveness, movement, growth, differentiation, and reproduction

# Homeostasis

- Homeostasis is a condition of equilibrium, or balance, in the body's internal environment
- Homeostasis is maintained by the body's regulatory processes
- Involves continuous monitoring and regulation of many factors
- Nervous and endocrine systems accomplish this communication via nervous impulses and hormones
  - Nervous system – fast acting; often short-lived effects
  - Endocrine – slow acting; often longer-lived effects

# Body Fluids & Homeostasis

- The survival of our body cells is dependent on the precise regulation of the chemical composition of their surrounding fluid
  - This fluid is known as extracellular fluid



# Control of Homeostasis

# Homeostasis

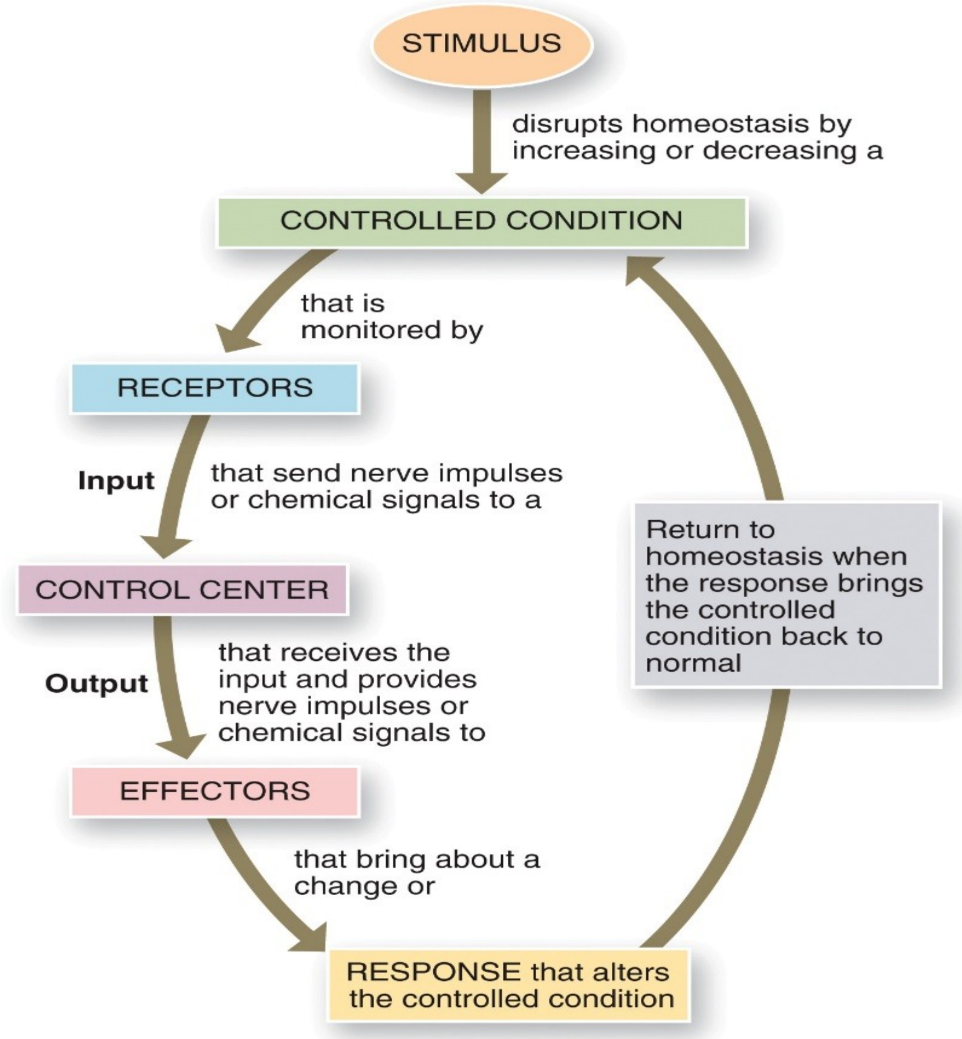
Homeostatic imbalances occur because of disruptions from the external or internal environments

Interactions Animation

- Communication, Regulation, and Homeostasis
- Negative Feedback Control of Temperature Regulation

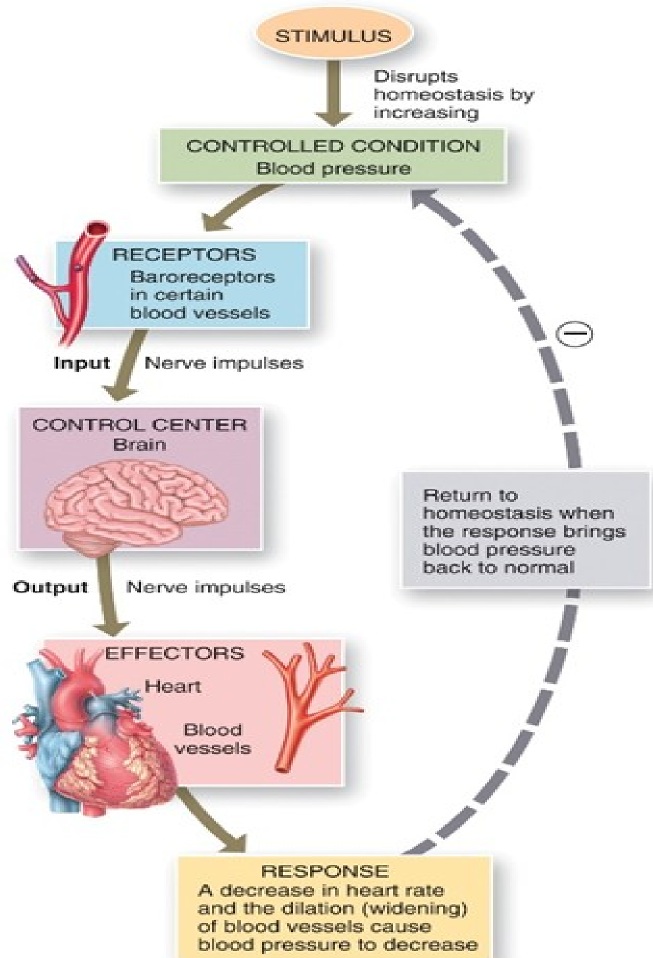
# Control of Homeostasis

The basic components of a feedback loop



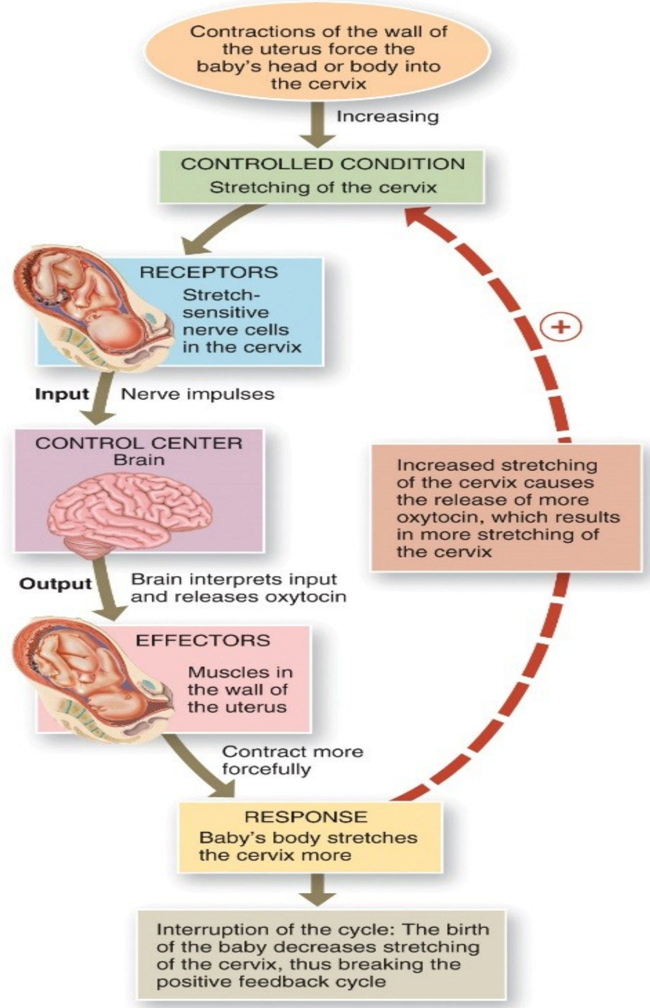
# Control of Homeostasis: Negative Feedback

- The response reduces or shuts off the original stimulus
- Examples:
  - Regulation of body temperature (a nervous mechanism)
  - Regulation of blood volume by ADH (an endocrine mechanism)



# Control of Homeostasis: Positive Feedback

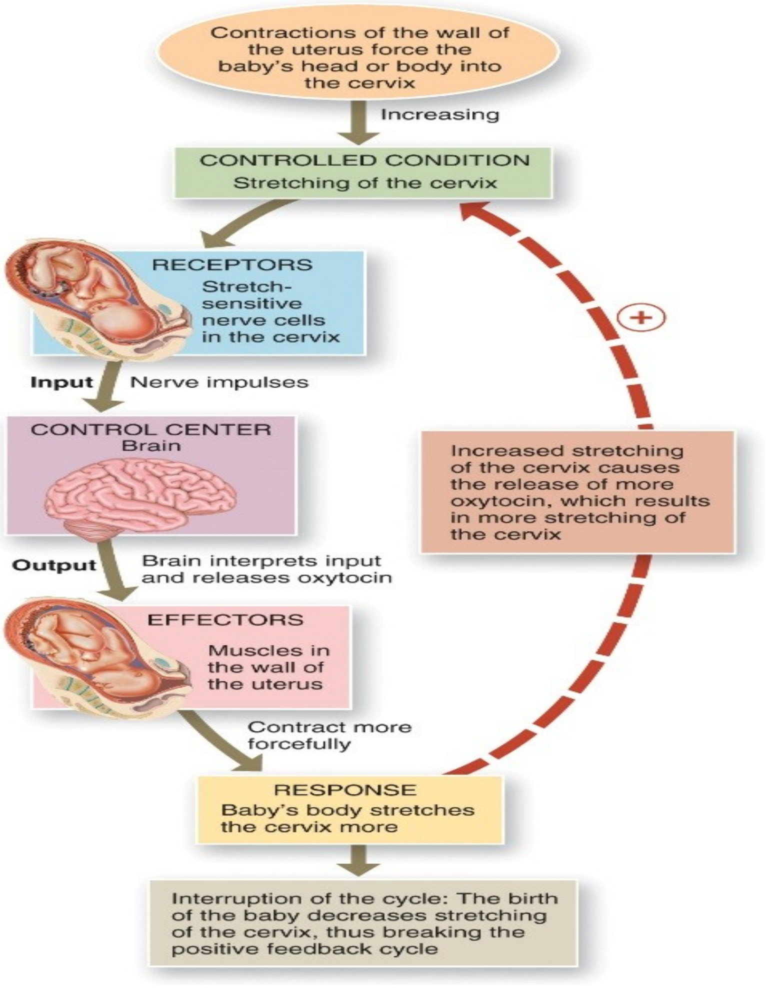
- Response enhances or exaggerates the original stimulus
- May exhibit a cascade or amplifying effect
- Usually controls infrequent events
  - Enhancement of labor contractions by oxytocin
  - Platelet plug formation and blood clotting



# Control of Homeostasis: Positive Feedback

## Interactions Animation:

- [Positive Feedback Control of Labor]



# Homeostatic Imbalances

- When homeostasis is disrupted it may result in disease, disorder, or even death
- Factors such as your genetic make-up, the air you breathe, and the food you eat can all affect your health

# Basic Anatomical Terminology

# Types of Anatomical Terminology

- Body positions
- Regional names
- Directional terms
- Planes and sections
- Body cavities

# Body Positions: The Anatomical Position

- Anatomical position is a standardized method of observing or imaging the body that allows precise and consistent anatomical reference

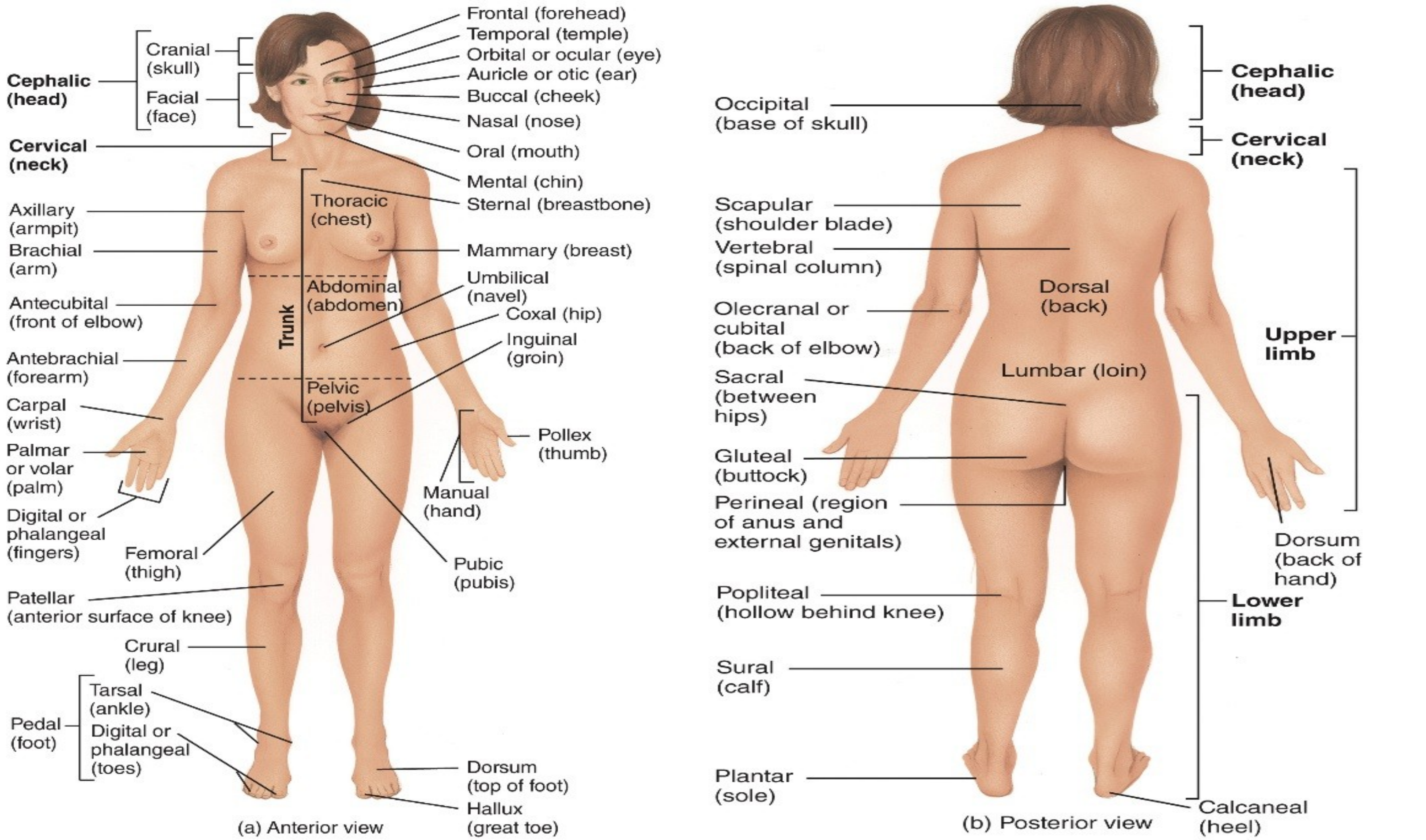
- Person stands erect, facing the observer, the upper extremities are places at the sides, the palms of the hands are turned forward, and the feet are flat on the floor

# Body Positions

- Two major divisions of body:
  - Axial
    - Head, neck, trunk
  - Appendicular
    - Limbs, shoulder girdle, pelvic girdle
- Anatomical terms to describe the reclining body are prone and supine
  - Prone = body lying facedown
  - Supine = body lying faceup

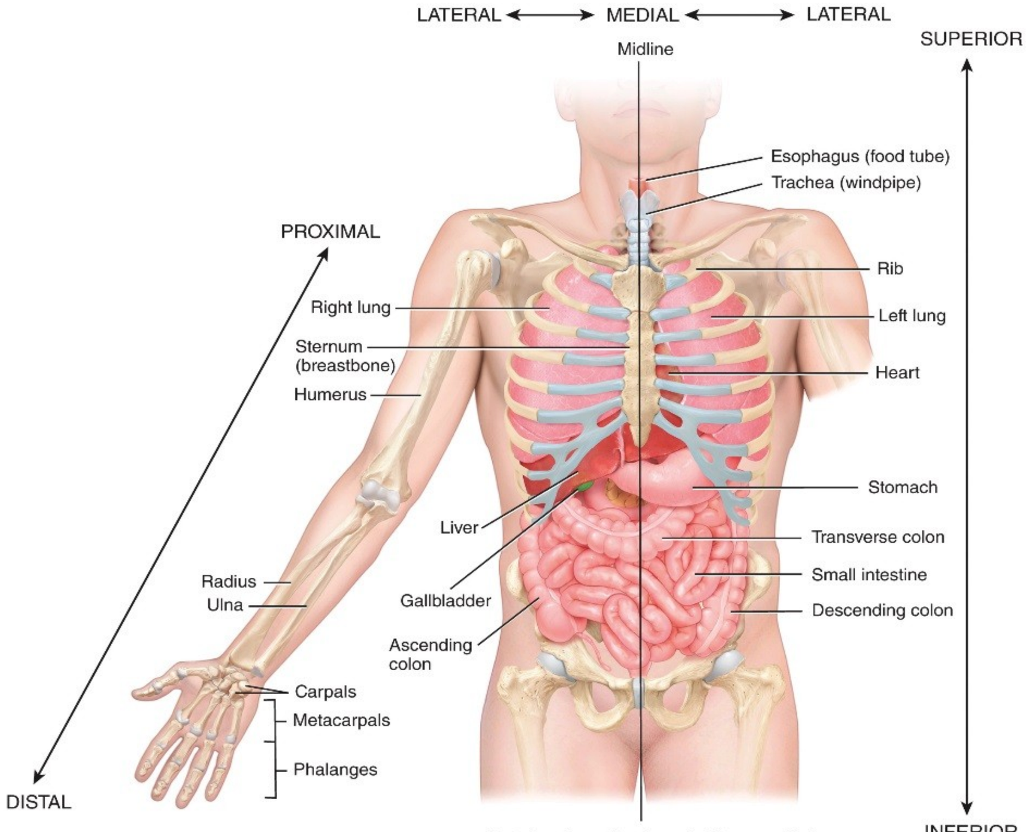
# Regional Names

- Names given to specific regions of the body for reference



# Directional Terms

- Used to precisely locate one part of the body relative to another



Anterior view of trunk and right upper limb

# Directional Terms

| Directional term | Definition | Example of Use |
|---|---|---|
| **Superior** (soo'-PĒR-ē-or) (*cephalic* or *cranial*) | Toward the head, or the upper part of a structure. | The heart is superior to the liver. |
| **Inferior** (in-FĒ-rē-or) (*caudal*) | Away from the head, or the lower part of a structure. | The stomach is inferior to the lungs. |
| **Anterior** (an-TĒR-ē-or) (*ventral*)* | Nearer to or at the front of the body. | The sternum (breastbone) is anterior to the heart. |
| **Posterior** (pos-TĒR-ē-or) (*dorsal*) | Nearer to or at the back of the body. | The esophagus (food tube) is posterior to the trachea (windpipe). |
| **Medial** (MĒ-dē-al) | Nearer to the midline (an imaginary vertical line that divides the body into equal right and left sides). | The ulna is medial to the radius. |

# Directional Terms

| Directional term | Definition | Example of Use |
|---|---|---|
| **Lateral** (LAT-er-al) | Farther from the midline. | The lungs are lateral to the heart. |
| **Intermediate** (in'-ter-MĒ-dē-at) | Between two structures. | The transverse colon is intermediate to the ascending and descending colons. |
| **Ipsilateral** (ip-si-LAT-er-al) | On the same side of the body as another structure. | The gallbladder and ascending colon are ipsilateral. |
| **Contralateral** (KON-tra-lat-er-al) | On the opposite side of the body from another structure. | The ascending and descending colons are contralateral. |
| **Proximal** (PROK-si-mal) | Nearer to the attachment of a limb to the trunk; nearer to the origination of a structure. | The humerus (arm bone) is proximal to the radius. |

# Directional Terms

| Directional term | Definition | Example of Use |
| --- | --- | --- |
| **Distal** (DIS-tal) | Farther from the attachment of a limb to the trunk; farther from the origination of a structure. | The phalanges (finger bones) are distal to the carpals (wrist bones). |
| **Superficial** (soo'-per-FISH-al) (*external*) | Toward or on the surface of the body. | The ribs are superficial to the lungs. |
| **Deep** (*Internal*) | Away from the surface of the body. | The ribs are deep to the skin of the chest and back. |

*Note that the terms *anterior* and *ventral* mean the same thing in humans. However, in four-legged animals ventral refers to the belly side and is therefore *inferior*. Similarly, the terms *posterior* and *dorsal* mean the same thing in humans, but in four-legged animals *dorsal* refers to the back side and is therefore *superior*.

# Planes and Sections

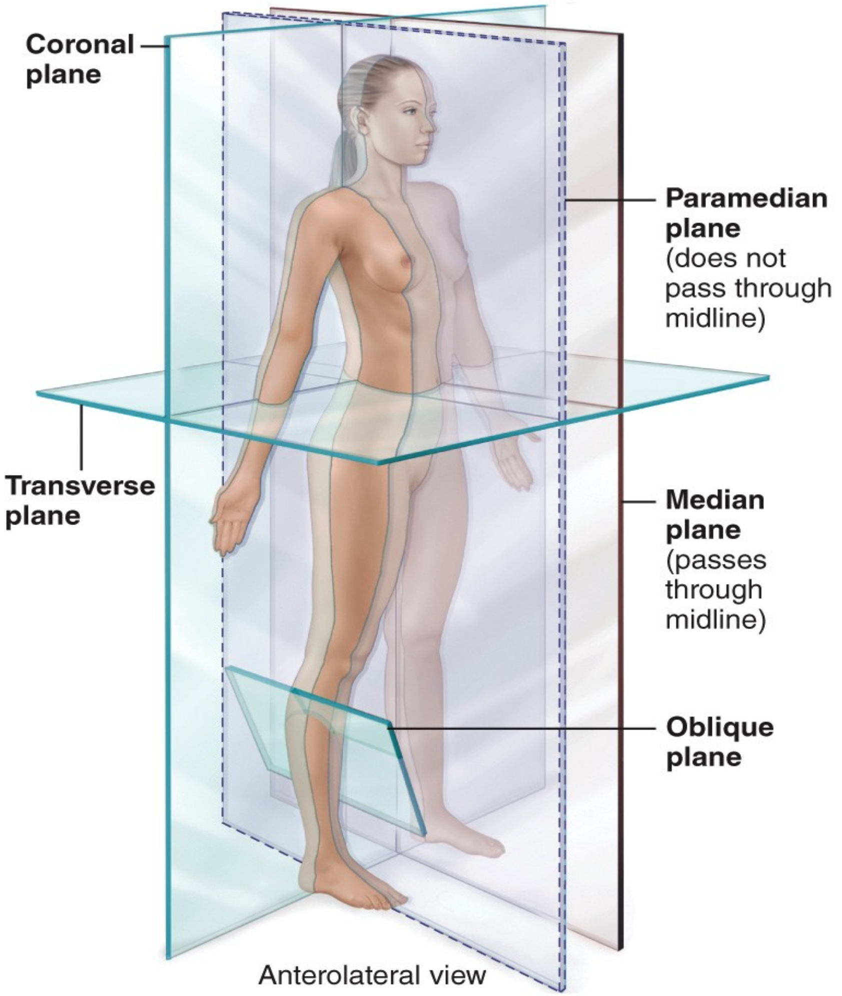
- Flat surface along which body structure is cut for anatomical study

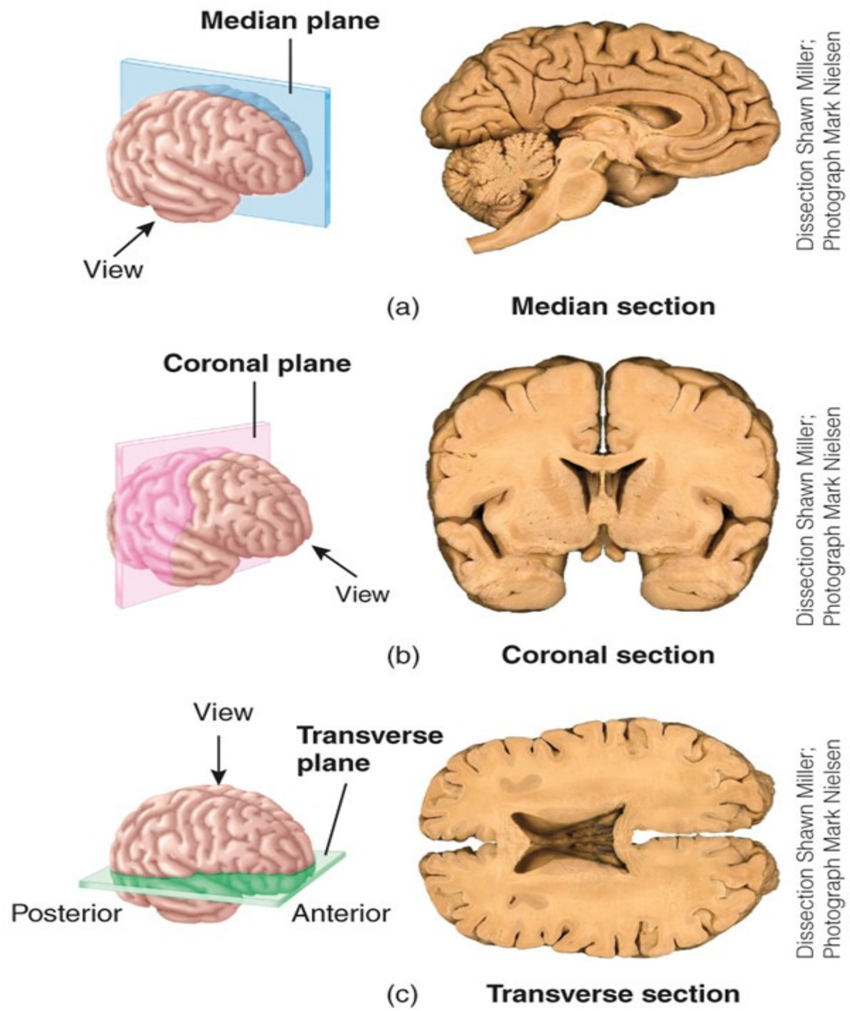
- Sagittal
  - Divides body vertically into left and right sides
    - Midsagittal/median
      - Lies on midline; equal left and right sides
    - Parasagittal/paramedian
      - Not on midline; unequal left and right sides

# Planes and Sections

- Frontal/coronal
  - Divides body vertically into anterior and posterior sections
- Transverse/horizontal
  - Divides body horizontally into superior and inferior sections
  - Produces a cross section
- Oblique
  - Cuts made diagonally

# Planes and Sections



# Body Cavities

- Dorsal cavity
  - Protects nervous system
  - Two subdivisions:
    - Cranial cavity
      - Encases brain
    - Vertebral cavity
      - Encases spinal cord

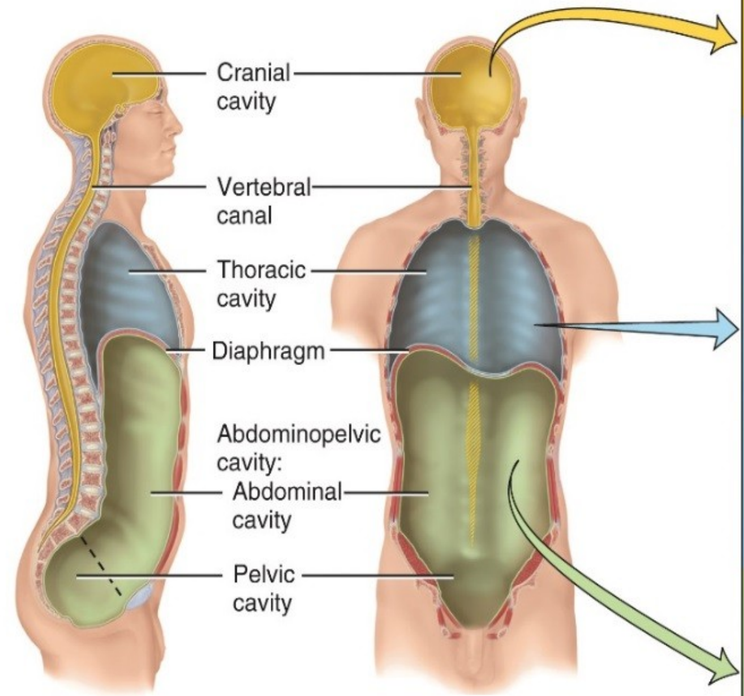
- Ventral cavity
  - Houses internal organs (viscera)
  - Two subdivisions (separated by diaphragm)
    - Thoracic cavity
    - Abdominopelvic cavity

# Ventral Cavities

- Thoracic cavity subdivisions:
  - Two pleural cavities
    - Each houses a lung
  - Mediastinum
    - Pericardial cavity
      - Encloses heart
    - Surrounds thoracic organs
      - Trachea, thyroid gland

- Abdominopelvic cavity subdivisions:
  - Abdominal cavity
    - Contains stomach, intestines, spleen, pancreas, liver
    - Most vulnerable to damage since it's not surrounded by bone
  - Pelvic cavity
    - Contains urinary bladder, reproductive organs, rectum

# Body Cavities



(a) Right lateral view
(b) Anterior view

| CAVITY | COMMENTS |
| --- | --- |
| **Cranial cavity** | Formed by cranial bones and contains brain. |
| **Vertebral canal** | Formed by vertebral column and contains spinal cord and the beginnings of spinal nerves. |
| **Thoracic cavity*** | Chest cavity; contains pleural and pericardial cavities and the mediastinum. |
| *Pleural cavity* | A potential space between the layers of the pleura that surrounds a lung. |
| *Pericardial cavity* | A potential space between the layers of the pericardium that surrounds the heart. |
| *Mediastinum* | Central portion of thoracic cavity between the lungs; extends from sternum to vertebral column and from first rib to diaphragm; contains heart, thymus, esophagus, trachea, and several large blood vessels. |
| **Abdominopelvic cavity** | Subdivided into abdominal and pelvic cavities. |
| *Abdominal cavity* | Contains stomach, spleen, liver, gallbladder, small intestine, and most of large intestine; the serous membrane of the abdominal cavity is the peritoneum. |
| *Pelvic cavity* | Contains urinary bladder, portions of large intestine, and internal organs of reproduction. |

*See **Figure 1.11** for details of the thoracic cavity.

# Serous Membranes

- Serous membranes are thin, double-layered membranes that cover the viscera within the thoracic and abdominal cavities and also line the walls of the thorax and abdomen
  - Visceral layer covers the internal organs
  - Parietal layer lines internal body walls
- Serous membranes reduce friction

# Serous Membranes in the Abdominal Cavity

Thoracic cavity

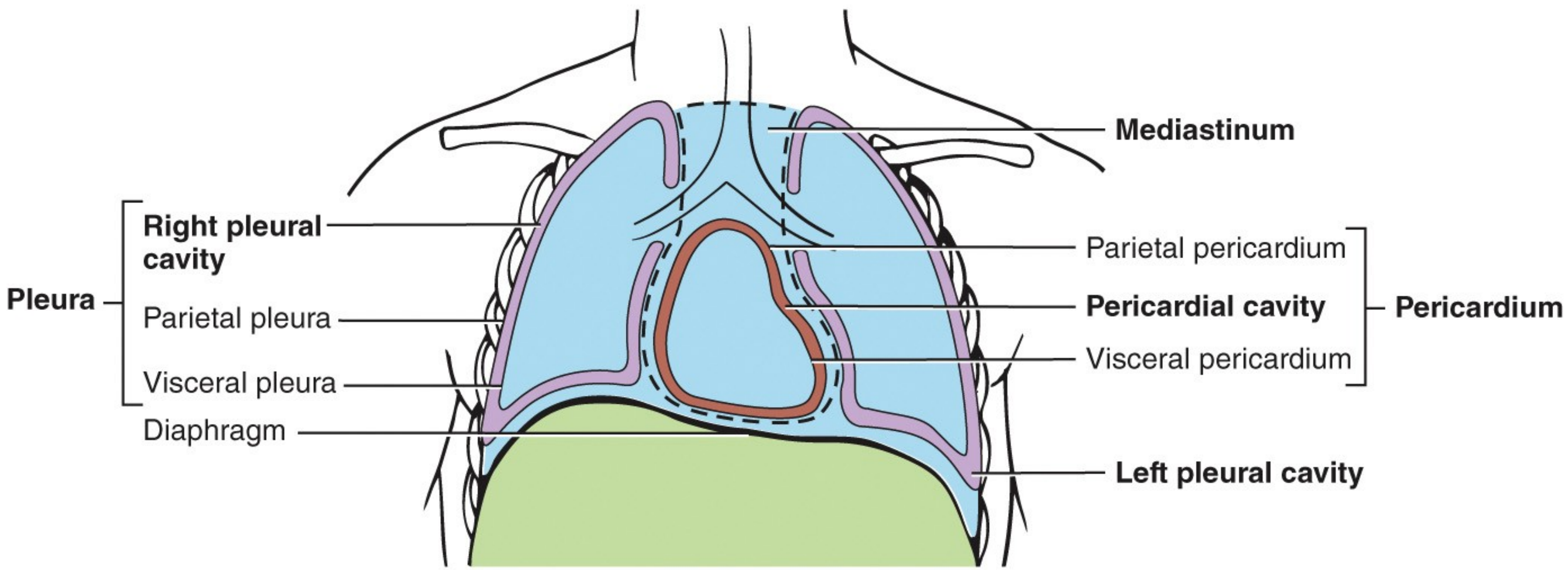
- Pericardium
  - Covers heart
- Pleura
  - Covers lungs

Abdominal cavity

- Peritoneum
  - Covers many abdominal organs
- Retroperitoneal
  - Classification given to some organs because they are not surrounded by peritoneum, rather, they are posterior to it

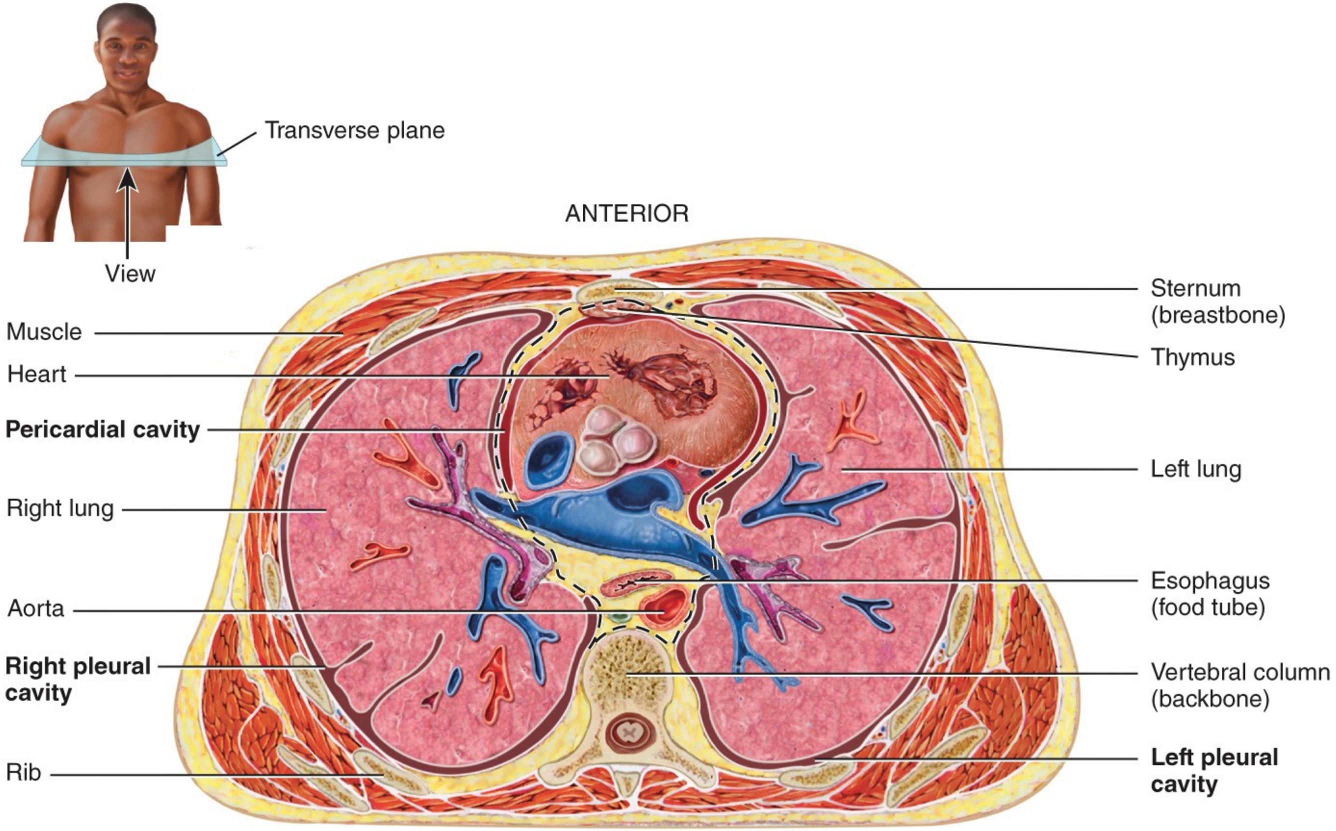
# Serous Membranes in the Thoracic Cavity

The pericardium and pleura cover the heart and lungs, respectively




(a) Anterior view of thoracic cavity

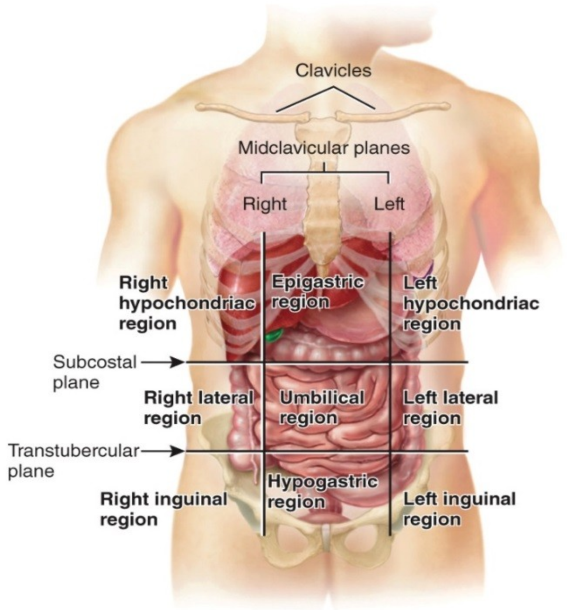
# Serous Membranes in the Thoracic Cavity



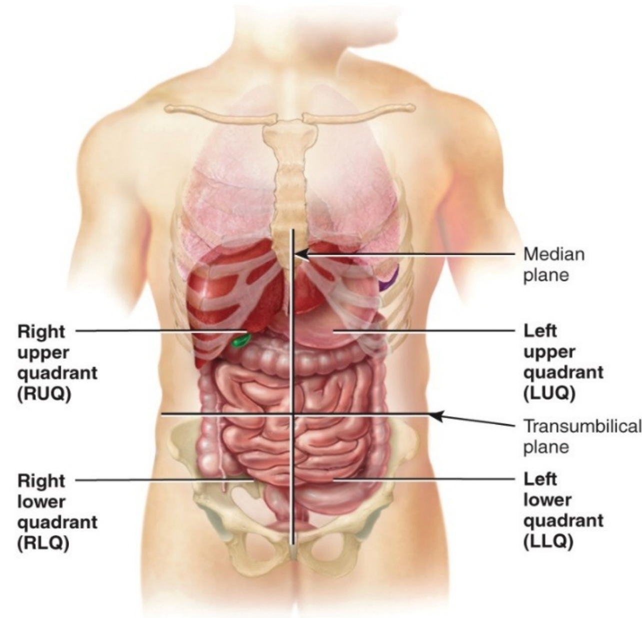

(b) Inferior view of transverse section of thoracic cavity

# Abdominopelvic Regions & Quadrants

- The abdominal cavity can be divided into regions or quadrants to easily describe the location of organs
  - “lateral” is often referred to as “lumbar”; “inguinal” is often referred to as “iliac”



(a) Anterior view showing location of abdominopelvic regions



(b) Anterior view showing location of abdominopelvic quadrants

# Aging and Homeostasis

- **Aging** is a normal process characterized by a progressive decline in the body's ability to restore homeostasis
  - Impacts all body systems
  - Produces structural and functional changes
  - Increases vulnerability to stress and disease

# Medical Imaging

- Medical imaging involves techniques that allow physicians to view images of the human body
- This allows physicians to diagnose anatomical and physiological abnormalities

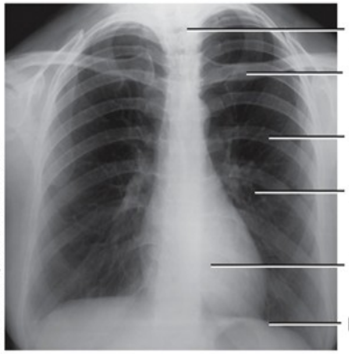
# Medical Imaging Procedures: Radiography is only one of a number of imaging procedures

## Table 1.3 Common Medical Imaging Procedures

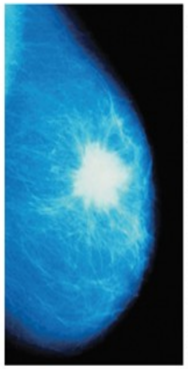
| Radiography |
| --- |
| ***Procedure***: A single barrage of x-rays passes through the body, producing an image of interior structures on x-ray-sensitive film. The resulting two- dimensional image is a radiograph (RA-dē-ō-graf'), commonly called an x-ray.<br>***Comments***: Relatively inexpensive, quick, and simple to perform; usually provides sufficient information for diagnosis. X-rays do not easily pass through dense structures, so bones appear white. Hollow structures, such as the lungs, appear black. Structures of intermediate density, such as skin, fat, and muscle, appear as varying shades of gray. At low doses, x-rays are useful for examining soft tissues such as the breast (**mammography**) and for determining bone density (**bone densitometry or DEXA scan**). It is necessary to use a substance called a contrast medium to make hollow or fluid-filled structures visible (appear white) in radiographs. X-rays make structures that contain contrast media appear white. The medium may be introduced by injection, orally, or rectally, depending on the structure to be imaged. Contrast x-rays are used to image blood vessels (**angiography**), the urinary system (**intravenous urography**), and the gastrointestinal tract (**barium contrast x-ray**). |

# Medical Imaging Procedures: Radiography

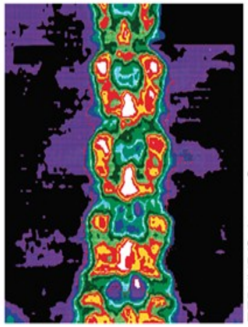
## Radiography



Radiograph of thorax in anterior view

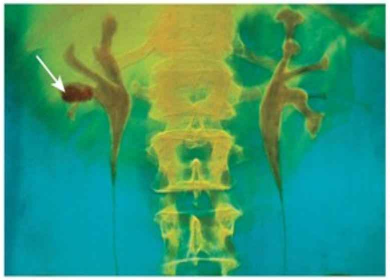


Mammogram of female breast showing cancerous tumor (white mass with uneven border)



Bone densitometry scan of lumbar spine in anterior view

Angiogram of adult human heart showing blockage in coronary artery (arrow)



Intravenous urogram showing kidney stone (arrow) in right kidney

Barium contrast x-ray showing cancer of the ascending colon (arrow)

# Medical Imaging Procedures: MRI and CT

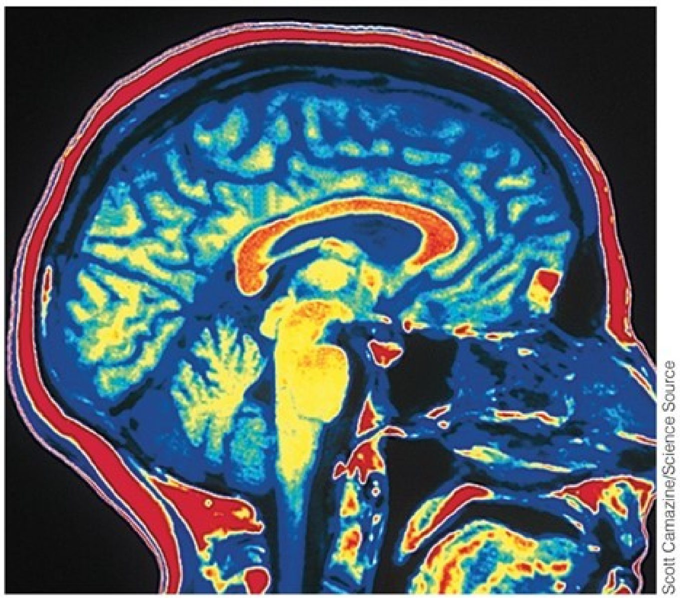
| Magnetic Resonance Imaging (MRI) | Computed Tomography (CT) |
|---|---|
| ***Procedure***: The body is exposed to a high-energy magnetic field, which causes protons (small positive particles within atoms, such as hydrogen) in body fluids and tissues to arrange themselves in relation to the field. Then a pulse of radio waves "reads" these ion patterns, and a color-coded image is assembled on a video monitor. The result is a two- or three-dimensional blueprint of cellular chemistry. | [formerly called computerized axial tomography (CAT) scanning)<br><br>***Procedure***: In this form of computer-assisted radiography, an x-ray beam traces an arc at multiple angles around a section of the body. The resulting transverse section of the body, called a CT scan, is shown on a video monitor. |

# Medical Imaging Procedures: MRI and CT

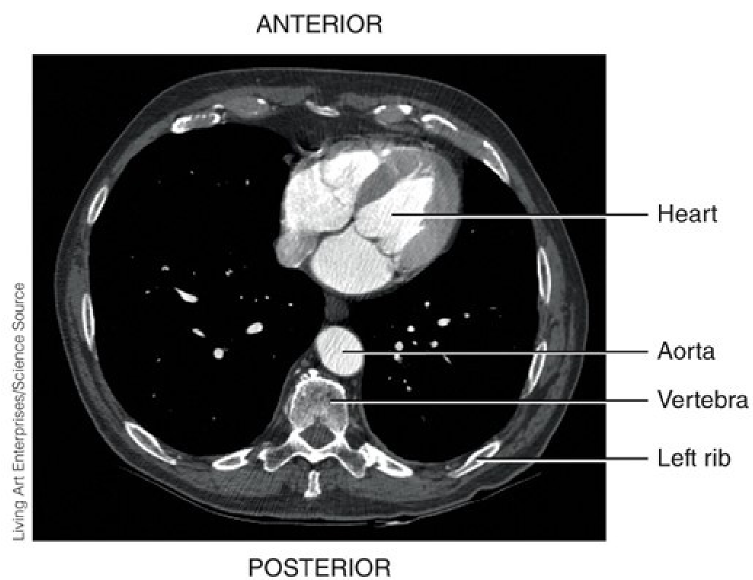
| Magnetic Resonance Imaging (MRI) | Computed Tomography (CT) |
|---|---|
| ***Comments***: Relatively safe but cannot be used on patients with metal in their bodies. Shows fine details for soft tissues but not for bones. Most useful for differentiating between normal and abnormal tissues. Used to detect tumors and artery-clogging fatty plaques; reveal brain abnormalities; measure blood flow; and detect a variety of musculoskeletal, liver, and kidney disorders. | **Comments:** Visualizes soft tissues and organs with much more detail than conventional radiographs. Differing tissue densities show up as various shades of gray. Multiple scans can be assembled to build three"−"dimensional views of structures (described next). Whole-body CT scanning typically targets the torso and appears to provide the most benefit in screening for lung cancers, coronary artery disease, and kidney cancers. |

# Medical Imaging Procedures: MRI and CT

| Magnetic Resonance Imaging (MRI) | Computed Tomography (CT) |
|---|---|




Magnetic resonance image of brain in sagittal section



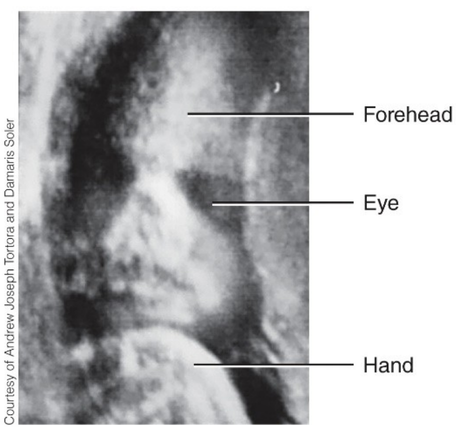

Computed tomography scan of thorax in inferior view

# Medical Imaging Procedures: Ultrasound Scanning

## Ultrasound Scanning

***Procedure***: High-frequency sound waves produced by a handheld wand reflect off body tissues and are detected by the same instrument. The image, which may be still or moving, is called a sonogram (SON-ō-gram) and is shown on a video monitor.

***Comments***: Safe, noninvasive, painless, and uses no dyes. Most commonly used to visualize the fetus during pregnancy. Also used to observe the size, location, and actions of organs and blood flow through blood vessels **(Doppler ultrasound).**






Sonogram of fetus

# Medical Imaging Procedures: CCTA and PET scan

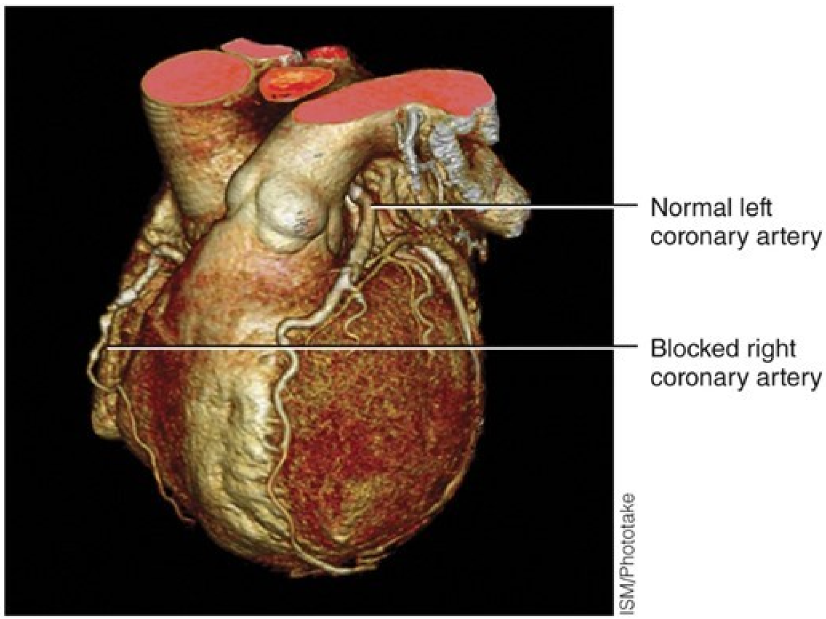
| Coronary (CARDIAC) Computed Tomography Angiography (CCTA) SCAN | Positron Emission Tomography (PET) |
|---|---|
| ***Procedure***: In this form of computer-assisted radiography, an iodine-containing contrast medium is injected into a vein and a beta blocker is given to decrease heart rate. Then, numerous x-ray beams trace an arc around the heart and a scanner detects the x-ray beams and transmits them to a computer, which transforms the information into a three-dimensional image of the coronary blood vessels on a monitor. The image produced is called a *CCTA scan* and can be generated in less than 20 seconds. | ***Procedure***: A substance that emits positrons (positively charged particles) is injected into the body, where it is taken up by tissues. The collision of positrons with negatively charged electrons in body tissues produces gamma rays (similar to x-rays) that are detected by gamma cameras positioned around the subject. A computer receives signals from the gamma cameras and constructs a *PET* scan image, displayed in color on a video monitor. The PET scan shows where the injected substance is being used in the body. In the PET scan image shown here, the black and blue colors indicate minimal activity; the red, orange, yellow, and white colors indicate areas of increasingly greater activity. |

# Medical Imaging Procedures: CCTA and PE T scan

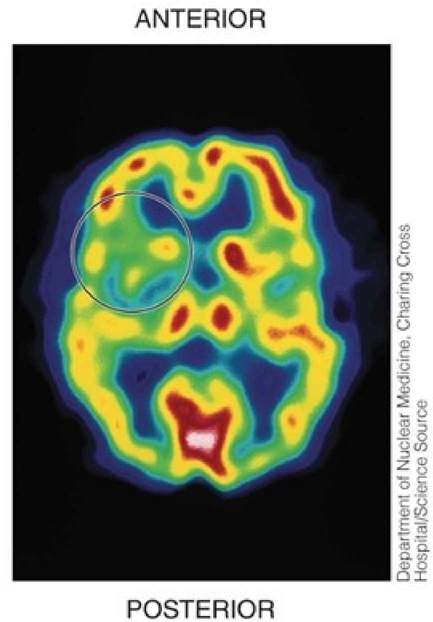
| Coronary (CARDIAC) Computed Tomography Angiography (CCTA) SCAN | Positron Emission Tomography (PET) |
|---|---|
| ***Comments***: Used primarily to determine if there are any coronary artery blockages (for example, atherosclerotic plaque or calcium) that may require an intervention such as angioplasty or stent. The CCTA scan can be rotated, enlarged, and moved at any angle. The procedure can take thousands of images of the heart within the time of a single heartbeat, so it provides a great amount of detail about the heart's structure and function. | ***Comments***: Used to study the physiology of body structures, such as metabolism in the brain or heart. |

# Medical Imaging Procedures: CCTA and PET scan

| Coronary (CARDIAC) Computed Tomography Angiography (CCTA) SCAN | Positron Emission Tomography (PET) |
| --- | --- |



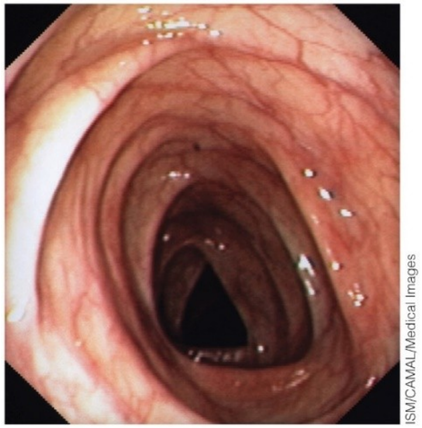

CCTA scan of coronary arteries




Positron emission tomography scan of transverse section of brain (circled area at upper left indicates where a stroke has occurred)

# Medical Imaging Procedures: Endoscopy

| Endoscopy |
| --- |
| ***Procedure***: Endoscopy involves the visual examination of the inside of body organs or cavities using a lighted instrument with lenses called an *endoscope*. The image is viewed through an eyepiece on the endoscope or projected onto a monitor.<br>***Comments***: Examples include *colonoscopy* (used to examine the interior of the colon, which is part of the large intestine), *laparoscopy* (used to examine the organs within the abdominopelvic cavity), and *arthroscopy* (used to examine the interior of a joint, usually the knee). |



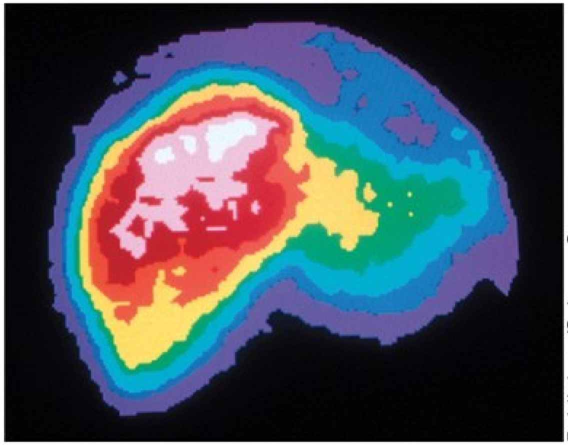

Interior view of colon as shown by colonoscopy

# Medical Imaging Procedures: Radionuclide Scanning

| Radionuclide Scanning |
| --- |
| ***Procedure*:** A *radionuclide* (radioactive substance) is introduced intravenously into the body and carried by the blood to the tissue to be imaged. Gamma rays emitted by the radionuclide are detected by a gamma camera outside the subject, and the data are fed into a computer. The computer constructs a *radionuclide image* and displays it in color on a video monitor. Areas of intense color take up a lot of the radionuclide and represent high tissue activity; areas of less intense color take up smaller amounts of the radionuclide and represent low tissue activity. **Single-photon-emission computed tomography (SPECT) scanning** is a specialized type of radionuclide scanning that is especially useful for studying the brain, heart, lungs, and liver. |
| ***Comments*:** Used to study activity of a tissue or organ, such as searching for malignant tumors in body tissue or scars that may interfere with heart muscle activity. |

# Medical Imaging Procedures: Radionuclide Scanning

**Radionuclide Scanning**



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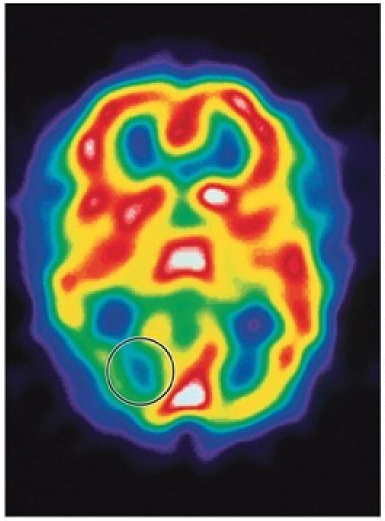
Radionuclide (nuclear) scan of normal human liver



Department of Nuclear Medicine, Charing Cross Hospital/Science Source

Single-photon-emission computed tomography (SPECT) scan of transverse section of the brain (the almost all green area at lower left indicates migraine attack)

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